

Revisiting Relative Clause Attachment by L1 Vietnamese-L2 English Bilinguals:
Effects of Prosody and Proficiency

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1. Background

Relative clause (RC) attachment ambiguity has been widely studied as a foundation for sentence processing and the interaction between syntax and prosody. In English, sentences such as (1) are structurally ambiguous because the RC *who has a smart dog* can attach either to the first noun phrase (NP1) *the brother* or to the second one (NP2) *the engineer*, also known as high attachment (HA) and low attachment (LA), respectively. Research on RC attachment has shown cross-linguistic differences in native speakers' default attachment preferences. English speakers generally prefer LA, whereas speakers of languages such as Spanish and French tend to show HA preferences (Cuetos & Mitchell, 1988; Fodor 2002).

(1) Jimmy met the brother of the engineer who has a smart dog.

One influential account proposes that attachment preferences are closely related to prosody. According to Fodor (1998, 2002), readers assign a default prosodic contour even during silent reading, and this implicit prosody guides syntactic parsing and interpretation. Prosodic phrasing can influence attachment preferences by signaling constituent boundaries during auditory processing. In particular, pauses and pitch resets may encourage listeners to group constituents differently, thus affecting whether the RC is interpreted as modifying NP1 or NP2. A prosodic break after NP1 may prompt more LA interpretations by grouping NP2 more closely with the RC, whereas a break after NP2 may favour HA interpretations by separating the RC from NP2. Although these effects have been reported for native speakers of English, less is known about how second language (L2) learners use prosodic information during online sentence interpretation.

This issue is somewhat interesting for Vietnamese-speaking learners of English because Vietnamese differs substantially from English in its RC structure and attachment patterns. Vietnamese RCs are relatively infrequent and are not commonly associated with the same type of attachment ambiguity found in English. As a result, Vietnamese learners may have less experience relying on syntactic ambiguity resolution strategies that are common in English sentence processing. Previous work has therefore raised the question of whether Vietnamese learners of English are sensitive to prosodic cues in ambiguous English RCs, and whether increased English proficiency improves the use of such cues.

In this writeup, I reanalyze data from an earlier auditory sentence interpretation experiment involving native English speakers and Vietnamese-speaking learners of English. Participants listened to ambiguous RC sentences produced under three prosodic conditions: no prosodic break, a break after NP1, and a break after NP2. After hearing each sentence, participants selected the intended referent of the RC. In addition to attachment responses, reaction times (RTs) were also collected and

analyzed in order to examine potential processing differences across conditions and participant groups.

The study addresses the following research questions:

- (i) Does prosodic phrasing affect RC attachment preferences in ambiguous English sentences?
- (ii) Do Vietnamese-speaking learners of English pattern differently from native speakers in their use of prosodic cues? Is English proficiency associated with their sensitivity to those cues?
- (iii) Do prosodic manipulations also affect processing time during attachment interpretation?

2. Methods

2.1. Dataset and Preprocessing

The dataset used in this analysis comes from a sentence interpretation experiment originally conducted as part of my undergraduate thesis project (Tran, 2023). The experiment investigated how prosodic phrasing affects the interpretation of structurally ambiguous RCs by native English speakers and Vietnamese learners of English. The three prosodic conditions (no break, break after NP1, and break after NP2) were designed to test whether prosodic phrasing biases listener toward HA or LA interpretations. Both attachment responses and RT measurements were included. Attachment responses were coded as binary outcomes indicating whether the participant selected the HA interpretation. RTs were recorded as the time between the display of the response options and the participant's selection.

The final dataset included 49 participants and 10 testing items. Participants were divided into three proficiency groups: Native speakers, Intermediate learners, and Advanced learners. Each observation in the dataset corresponded to a single participant response to one stimulus item under one prosodic condition.

The primary dependent variables analyzed in this study are:

- `attachment_HA`: binary attachment responses (1 = HA, 0 = LA);
- `logRT`: log-transformed reaction times.

The main predictor variables are:

- `condition`: prosodic condition (*No break, Break after NP1, Break after NP2*);
- `proficiency_level`: participant language group and/or proficiency (native English vs. Vietnamese-speaking intermediate learner vs. advanced learner);
- `cloze_z`: standardized English proficiency/cloze test score (for learner analysis).

Random intercepts were included for both `participant` and `item` in the mixed-effects models.

Data preprocessing was carried out in R (R Core Team, 2026) prior to exploratory analysis and model fitting. The original dataset included both target items and filler trials, as well as raw response and RT data. Only target experimental items were retained for the current analyses. Several preprocessing steps were carried out to prepare the data for mixed-effects modeling. First, attachment responses

were recoded into a binary variable representing HA interpretations: HA responses were coded as 1 and LA responses as 0. This coding allows attachment preferences to be analyzed using logistic mixed-effects regression. Next, reaction times were cleaned and transformed. Since RTs are typically positively skewed, raw RTs were log-transformed to improve normality and interpretability. This also reduces the influence of extremely long responses.

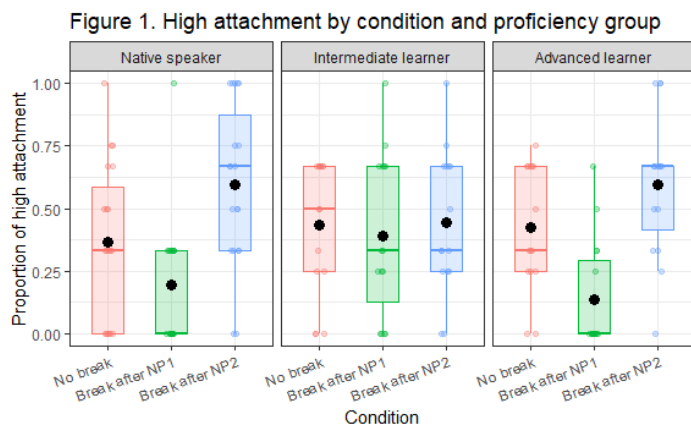
In addition, RTs were standardized within participants to account for baseline differences in individual response time. This participant-centered standardized RT measure (RT_z_part) was used to identify potential outliers. Extreme RT observations were removed based on within-participant standardized RT values. Following inspection of the RT distribution and exploratory analyses, observations with $|z| > 2.5$ were excluded from the RT analyses. This trimming procedure removed a small number of observations (3) while still preserving the overall pattern of the dataset. Attachment analyses retained all valid categorical responses, whereas RT analyses were conducted on the trimmed dataset only. The full dataset contained 490 observations for the attachment analyses and 487 observations for the RT analyses after RT trimming.

The dataset also contains a continuous proficiency measure based on cloze test scores for the learner group. Cloze scores were standardized/z-scored prior to modeling in order to reduce scaling issues in interaction models.

2.2. EDA

Exploratory data analysis was conducted prior to model fitting to examine overall response patterns and assess whether the observed trends aligned with the research questions. The EDA focuses on two main dependent variables: attachment responses and RTs. Furthermore, I also examined the relationship between learner proficiency and attachment preferences.

Figure 1 presents the proportion of HA responses across prosodic conditions and proficiency groups. Each point represents a participant's mean HA rate within a condition, while the black dots indicate group means. Across all groups, the *Break after NP2* condition shows the highest HA rates, suggesting that a prosodic boundary after NP2 biased listeners toward attaching the RC to NP1. In contrast, the *Break after NP1* condition generally produces lower HA rates, consistent with stronger low LA interpretations. The *No break* condition falls between the two prosodically manipulated conditions.



Differences between proficiency groups are also visible. Native English speakers show the clearest distinction between conditions, with relatively high HA rates in the *Break after NP2* condition and lower HA rates elsewhere. Advanced learners pattern similarly to native English speakers, though with greater variability across participants. Intermediate learners show weaker separation between conditions, suggesting lower sensitivity to prosodic cues in detecting attachment types overall.

Figure 2 displays mean log-transformed RTs across conditions and proficiency groups. In general, RT differences between conditions appeared smaller than the differences observed for attachment responses. However, some trends are still clear to see. Native English speakers show relatively fast responses in the *Break after NP1* condition and slower responses in the *Break after NP2* condition, possibly reflecting increased processing demands associated with the HA interpretations. Advanced learners display relatively stable RTs across conditions, whereas intermediate ones show a slight decrease in RTs from the *No break* condition to the *Break after NP1* and the *Break after NP2* condition. Error bars indicate substantial variability, particularly among learner groups, indicating that any RT effects might be weaker or less consistent than attachment effects.

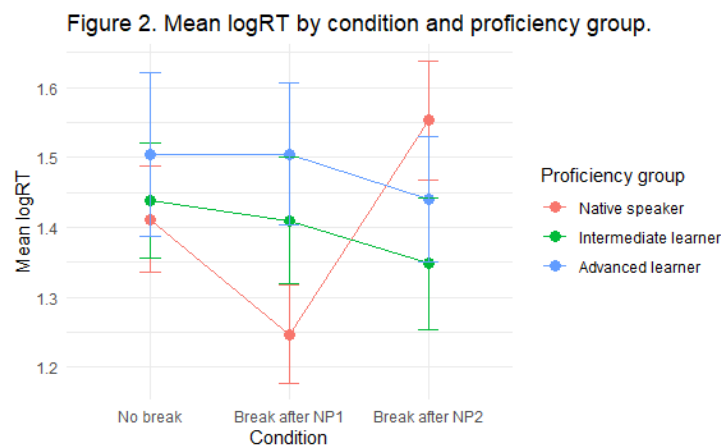
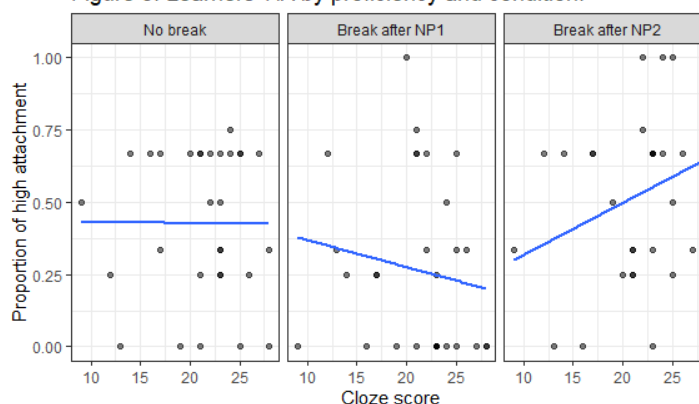


Figure 3 explores the relationship between learner proficiency and HA responses across conditions. Because proficiency scores are only meaningful for Vietnamese learners of English, native English speakers were excluded from this analysis. The scatterplots below suggest that the mentioned relationship varies across prosodic conditions. In the *No break* condition, HA rates appear relatively stable across proficiency scores. In the *Break after NP1* condition, higher proficiency scores are weakly associated with lower HA rates, corresponding to stronger LA interpretations. In contrast, the *Break after NP2* condition shows a positive relationship between cloze scores and HA responses, meaning that the more proficient learners may be more sensitive to prosodic cues favouring HA interpretations.

Figure 3. Learners' HA by proficiency and condition.



Overall, the EDA suggests that prosodic phrasing influences attachment interpretation. The exploratory plots also indicate potential proficiency-related differences in the use of prosodic information, motivating the inclusion of participant language group and proficiency as possible predictors in subsequent mixed-effects analyses.

3. Models and Results

Categorical predictors are contrast-coded to make the model outputs easier to interpret. For the prosodic condition variable, centered sum coding is used instead of the default reference-level coding in R. With this coding method, the model intercept represents the overall average across all conditions, while each coefficient shows how a condition differs from that overall average. Participant group predictors were also coded using centered contrasts in models involving categorical proficiency levels (*Native speaker*, *Intermediate learner*, *Advanced learner*).

Table 1. Contrast coding scheme for categorical predictors.

```
> contrasts(target_rc$condition)
      [,1] [,2]
No break      1      0
Break after NP1  0      1
Break after NP2 -1     -1
> contrasts(target_rc$proficiency_level)
      [,1] [,2]
Native speaker      1      0
Intermediate learner  0      1
Advanced learner    -1     -1
```

3.1. Attachment model

The attachment analysis is carried out using a generalized linear mixed-effects model (GLMM) with a binomial distribution. The dependent variable, `attachment_HA`, is binary: responses are coded as 1 for HA and 0 for LA. The model formula is stated in (2). The fixed effects include the three prosodic conditions and the three participant proficiency groups. The interaction term tests whether the effect of prosodic phrasing differs across proficiency levels. Random intercepts for participant and item are included to account for repeated observations from the same individuals and stimulus items.

(2) `attachment_HA ~ condition * proficiency_level + (1 | participant) + (1 | item)`

Table 2 presents the results of the GLMM predicting HA responses from prosodic condition, proficiency level, and their interaction. The model revealed a significant effect associated with the *Break after NP1* contrast ($\beta = -0.88$, $p < 0.001$). The negative coefficient indicates that participants produced fewer HA responses in this condition overall. This pattern is consistent with Figure 1, which shows that *Break after NP1* elicits the lowest HA rates. In contrast, *No break* is not statistically significant ($\beta = -.11$, $p = 0.440$), suggesting that HA responses in this condition do not differ substantially from the overall mean across conditions.

The main effects of proficiency level are also not significant. Neither the *Native speaker* contrast ($\beta = -0.05$, $p = 0.791$) nor the *Intermediate learner* ($\beta = 0.15$, $p = -.448$) differs significantly from the overall proficiency mean. This suggests that the three proficiency groups do not show strong overall differences in HA preference.

Most interaction terms between condition and proficiency level are not statistically significant, indicating that the effect of prosody is generally similar across groups. However,, the interaction between the *Break after NP1* contrast and the *Intermediate learner* is statistically significant ($\beta = 0.71$, $p=0.001$). This result means that *Intermediate learners* pattern differently from the overall proficiency mean specifically in the *Break after NP1* condition. In Figure 1, *Intermediate learners* appear to show weaker differentiation between prosodic conditions than *Native speakers* and *Advanced learners*.

Random-effects estimates also indicate variability across both participants and items, supporting the inclusion of both grouping factors in the model structure. Participant-level variance ($\tau_{00} = 0.42$) was somewhat larger than item-level variance ($\tau_{00} = 0.34$), suggesting that individual differences between listeners contributed more variability to attachment behavior than differences between stimulus items. Furthermore, the conditional R^2 value (0.301) indicates that including participant-level and item-level random effects substantially improved the model's ability to account for variability in attachment responses.

3.2. RT model

The RT analysis is carried out using a linear mixed-effects model (LMM). Log-transformed RTs are used as the dependent variable. The model formula is stated in (3). The fixed effects include the three prosodic conditions and the three participant proficiency groups. The interaction term tests whether the effect of prosodic phrasing on processing time differs across proficiency levels. Random intercepts for participant and item are also included.

Table 2. GLMM predicting HA responses.

Predictors	HA responses			
	Log-Odds	std. Error	Statistic	p
(Intercept)	-0.56	0.23	-2.37	0.018
No break contrast	0.11	0.15	0.77	0.440
Break after NP1 contrast	-0.88	0.17	-5.32	<0.001
Native speaker contrast	-0.05	0.19	-0.26	0.791
Intermediate learner contrast	0.15	0.20	0.76	0.448
No break × Native speaker	-0.11	0.20	-0.53	0.593
Break after NP1 × Native speaker	-0.17	0.22	-0.77	0.440
No break × Intermediate learner	-0.07	0.21	-0.33	0.739
Break after NP1 × Intermediate learner	0.71	0.22	3.24	0.001
Random Effects				
σ^2	3.29			
τ_{00} participant	0.42			
τ_{00} item	0.34			
ICC	0.19			
N participant	49			
N item	10			
Observations	490			
Marginal R^2 / Conditional R^2	0.138 / 0.301			

(3) $\text{logRT} \sim \text{condition} * \text{proficiency_level} + (1 | \text{participant}) + (1 | \text{item})$

Table 3 presents the results of the LMM formula from above. Overall, the RT model shows weaker effects than the attachment model, and none of the fixed effects reaches conventional significance levels. However, the *Break after NP1* condition approaches significance ($\beta = -0.17, p = 0.097$), suggesting faster responses in this condition relative to the baseline *No break* condition. This pattern is consistent with Figure 2, where Native speakers show noticeably lower mean logRTs in the *Break after NP1* condition.

No significant main effects of proficiency level are observed, indicating that the three groups do not differ substantially in overall processing speed. Similarly, none of the interaction terms reaches significance, although the negative coefficients for the *Break after NP2* interactions suggest a possible tendency for learner groups to respond faster than Native speakers in this condition. This pattern can also be seen in Figure 2, where Native speakers show slower RTs in the *Break after NP2* condition when compared to the two learner groups.

The random-effects estimates indicated greater variability across participants than across items, suggesting that individual differences between listeners contributed more to RT variation than differences between stimulus items.

3.3. Learner-only model

I use a GLMM to carry out the learner-only attachment analysis. The model formula is stated in (4). The fixed effects include prosodic condition, standardized cloze score (*cloze_z*), and their interaction. The interaction term tests whether the effect of prosodic phrasing varies as a function of learner proficiency. Random intercepts for participant and item are included.

(4) $\text{attachment_HA} \sim \text{condition} * \text{cloze_z} + (1 | \text{participant}) + (1 | \text{item})$

Table 3. LMM predicting logRTs.

<i>Predictors</i>	logRT	
	<i>Estimates</i>	<i>p</i>
(Intercept)	1.41	<0.001
condition [Break after NP1]	-0.17	0.097
condition [Break after NP2]	0.15	0.141
proficiency level [Intermediate learner]	0.02	0.880
proficiency level [Advanced learner]	0.09	0.580
condition [Break after NP1] × proficiency level [Intermediate learner]	0.12	0.418
condition [Break after NP2] × proficiency level [Intermediate learner]	-0.22	0.153
condition [Break after NP1] × proficiency level [Advanced learner]	0.18	0.230
condition [Break after NP2] × proficiency level [Advanced learner]	-0.22	0.156
Random Effects		
σ^2	0.32	
τ_{00} participant	0.11	
τ_{00} item	0.00	
ICC	0.27	
$N_{\text{participant}}$	49	
N_{item}	10	
Observations	487	
Marginal R^2 / Conditional R^2	0.019 / 0.287	

Table 4 presents the results of this model. It reveals a significant effect associated with the *Break after NP1* contrast ($\beta = -0.71$, $p < .001$). This pattern is consistent with the exploratory plots, where learners generally showed lower HA rates in the *Break after NP1* condition. The main effect of cloze score was not statistically significant ($\beta = 0.07$, $p = 0.694$), suggesting that proficiency alone did not reliably predict overall attachment preferences across conditions.

The interaction between the *Break after NP1* contrast and cloze score was statistically significant ($\beta = -0.40$, $p = 0.043$). This result suggests that the effect of prosodic condition varied depending on learner proficiency. Specifically, higher cloze scores were associated with fewer HA responses in the *Break after NP1* condition, indicating that more proficient learners showed stronger LA preferences when prosody favoured that interpretation. In contrast, the interaction between the *No break* contrast and cloze score was not significant ($p = 0.931$), suggesting that proficiency did not substantially affect attachment preferences in the absence of strong prosodic cues.

Finally, the random-effects estimates showed meaningful variability across both participants and items ($\tau_{00} = 0.34$ for both grouping factors), suggesting that both individual differences between learners and differences between stimulus items contributed to variability in attachment responses.

4. Model validation

Model diagnostics were conducted to assess whether the mixed-effects models met the relevant assumptions. All three models were checked for singularity, and none were found to be singular.

Table 4. Learner-only LMM predicting HA responses with cloze scores as a continuous predictor,

Predictors	Learner HA responses			
	Log-Odds	std. Error	Statistic	p
(Intercept)	-0.50	0.25	-2.01	0.045
No break contrast	0.14	0.18	0.77	0.440
Break after NP1 contrast	-0.71	0.20	-3.63	<0.001
Cloze score	0.07	0.17	0.39	0.694
No break $\times$ Cloze score	-0.02	0.18	-0.09	0.931
Break after NP1 $\times$ Cloze score	-0.40	0.20	-2.02	0.043
Random Effects				
σ^2	3.29			
τ_{00} participant	0.34			
τ_{00} item	0.34			
ICC	0.17			
N _{participant}	30			
N _{item}	10			
Observations	300			
Marginal R ² / Conditional R ²	0.093 / 0.247			

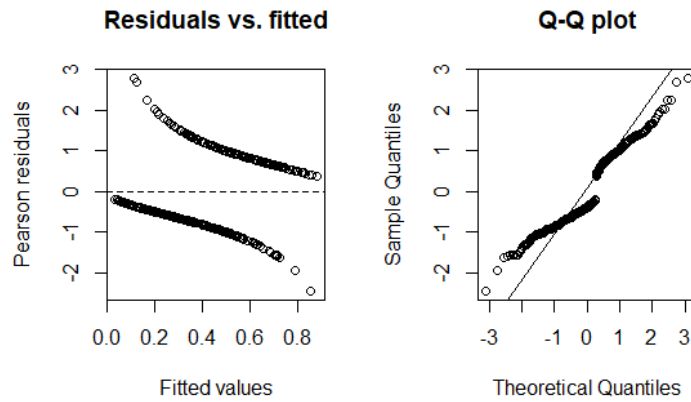


Figure 4. Residual plots for the attachment model.

Inspection of the Pearson residual plots for the attachment model from Figure 4 shows the expected banding pattern associated with binary response data. The Q-Q plot deviated somewhat from normality.

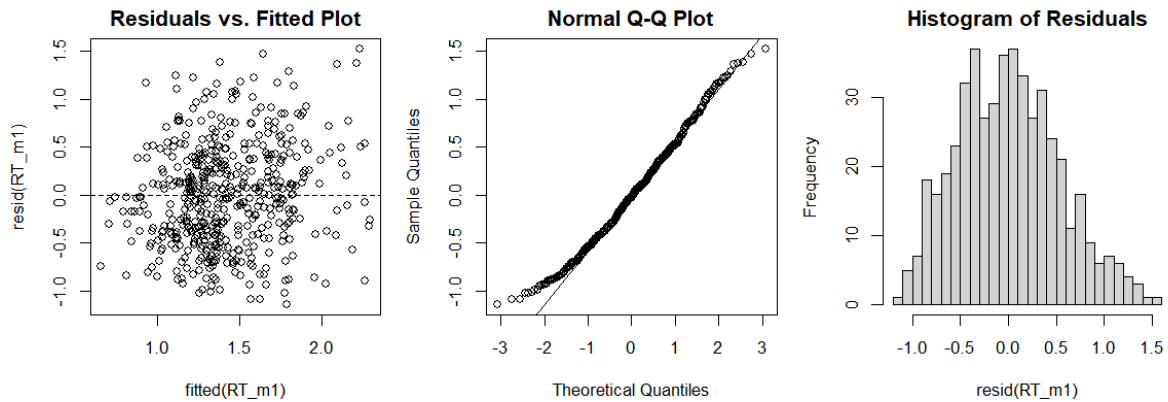


Figure 5. Residual plots for the RT model.

For the RT LMM, residual diagnostics from Figure 5 suggested that the model assumptions were reasonably satisfied after log transformation of RTs. The residuals appeared approximately normally distributed, and the Q-Q plot showed only minor deviations from the reference line. In addition, the residuals-versus-fitted plot did not show strong systematic patterns. Overall, these diagnostics suggest that the RT model provided an adequate fit to the data.

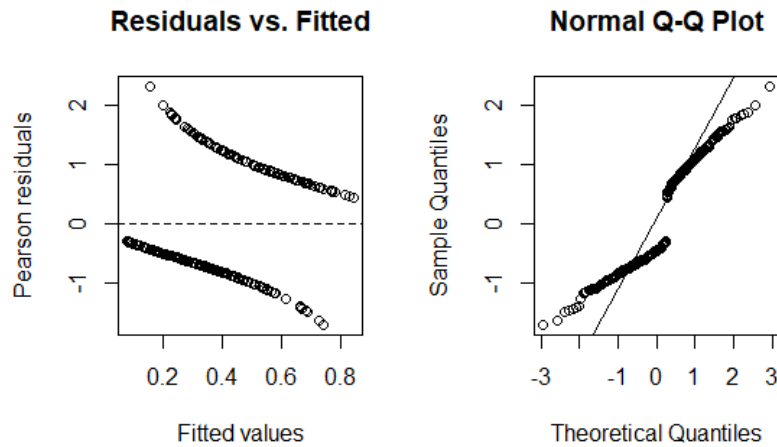


Figure 6. Residual plots for the learner-only model.

Figure 6 shows validation results for the learner-only GLMM. As in the main attachment model, the Pearson residuals exhibit the characteristic banding pattern commonly associated with binary response data. Some deviations from normality are visible in the Q-Q plot, but these are expected in binomial mixed-effects models. Overall, the diagnostic plots do not reveal strong evidence of model misspecification, suggesting that the learner-only model provides an adequate fit to the data.

5. Discussion and Conclusion

This writeup investigates how prosodic phrasing influences RC attachment interpretation in English, with particular attention to Vietnamese-speaking learners of English and the role of proficiency. The analyses address three main research questions concerning attachment preferences, learner sensitivity to prosodic cues, and processing time.

The first research question asks whether prosodic phrasing affects RC attachment preferences in ambiguous English sentences. The results from both exploratory analyses and the mixed-effects models provide evidence that prosodic phrasing does influence attachment interpretation. Across analyses, the *Break after NP1* condition is associated with fewer HA responses, indicating stronger LA preferences in this condition. In contrast, the *Break after NP2* condition generally elicits higher HA rates. These findings support previous research that suggests that prosodic boundaries guide syntactic parsing by signaling constituent grouping during sentence processing.

The second research question asks whether Vietnamese-speaking learners of English pattern differently from native English speakers in their use of prosodic cues, and whether English proficiency is linked to sensitivity to those cues. The attachment model does not reveal strong overall group differences, as most interaction terms between condition and proficiency level are not statistically significant. However, the exploratory plots show noticeable differences between groups. Native English speakers display the clearest differentiation across prosodic conditions, while intermediate learners show weaker separation between conditions. Advanced learners generally pattern more similarly to native English speakers, especially in the *Break after NP2* condition.

More importantly, the learner-only model reveals a significant interaction between the *Break after NP1* condition and cloze score. Higher proficiency learners show fewer HA responses in this condition, indicating stronger LA preferences when prosody favoured that interpretation. This suggests that English proficiency is associated with greater sensitivity to prosodic cues during ambiguity resolution. Together, these findings indicate that Vietnamese-speaking learners are able to use prosodic information during RC attachment processing, although sensitivity to these cues appears to strengthen with increasing proficiency.

The third research question concerns whether prosodic manipulations affect processing time during attachment interpretation. Compared to the attachment models, the RT model shows weaker effects overall: none of the fixed effects reaches conventional significance levels, although the *Break after NP1* condition approaches significance and is associated with faster responses. This trend is also visible in the exploratory plots, particularly among native English speakers. The lack of strong RT effects suggests that prosodic phrasing may have influences interpretation preferences more consistently than processing speed. It is also possible that the RT data contained greater individual variability, reducing the model's ability to detect smaller effects.

Several limitations should also be considered. First, the number of experimental items was relatively small, which may have limited statistical power, especially for interaction effects. Second, learner groups show considerable variability in both attachment behaviour and RTs. Future work could incorporate additional participant-level variables such as age of acquisition or amount of spoken English exposure. Finally, the present analyses focus on offline attachment choices and overall RTs. Future studies could use online processing measures such as eye-tracking to examine the time course of prosodic disambiguation more directly.

Despite these limitations, the study demonstrates that prosodic phrasing contributes to RC attachment interpretation in English and that Vietnamese-speaking learners are sensitive to these cues to varying degrees. In particular, the learner-only model suggests that increased English proficiency is associated with greater use of prosodic information during ambiguity resolution. Overall, the findings contribute to broader research on bilingual sentence processing and the interaction between syntax, prosody, and second language acquisition.

6. References

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